

FINAL PROJECT S14 • DATA ANALYSIS

The Importance of AI in Our Lives and Work

An exploratory analysis of the impact of Artificial Intelligence on the global labor market (2020–2026)

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Dataset: Future of Work in the Age of AI (2020–2026) • Kaggle

Why study AI's impact on work?

Artificial Intelligence is no longer a futuristic promise — it has become part of everyday professional life. But the real impact varies a lot — depending on occupation, industry, and country.



Unequal risk

Not every occupation is exposed to automation the same way.



Salary impact

Is adopting AI really linked to earning more (or less)?



Global difference

Countries and industries respond in very different ways.

5 Research Questions

- 1 Which occupations and industries are most at risk of being replaced by AI?
- 2 Is AI adoption related to salary increases or decreases?
- 3 How has automation risk evolved from 2020 to 2026?
- 4 Is there a difference between countries in AI adoption level and salary impact?
- 5 Which occupations have the greatest reskilling urgency?

THE DATASET

Future of Work in the Age of AI

Synthetic, realistic data on the impact of AI on the global labor market, available on Kaggle.

15,000

Records

8

Industries

9

Countries

10

Occupations

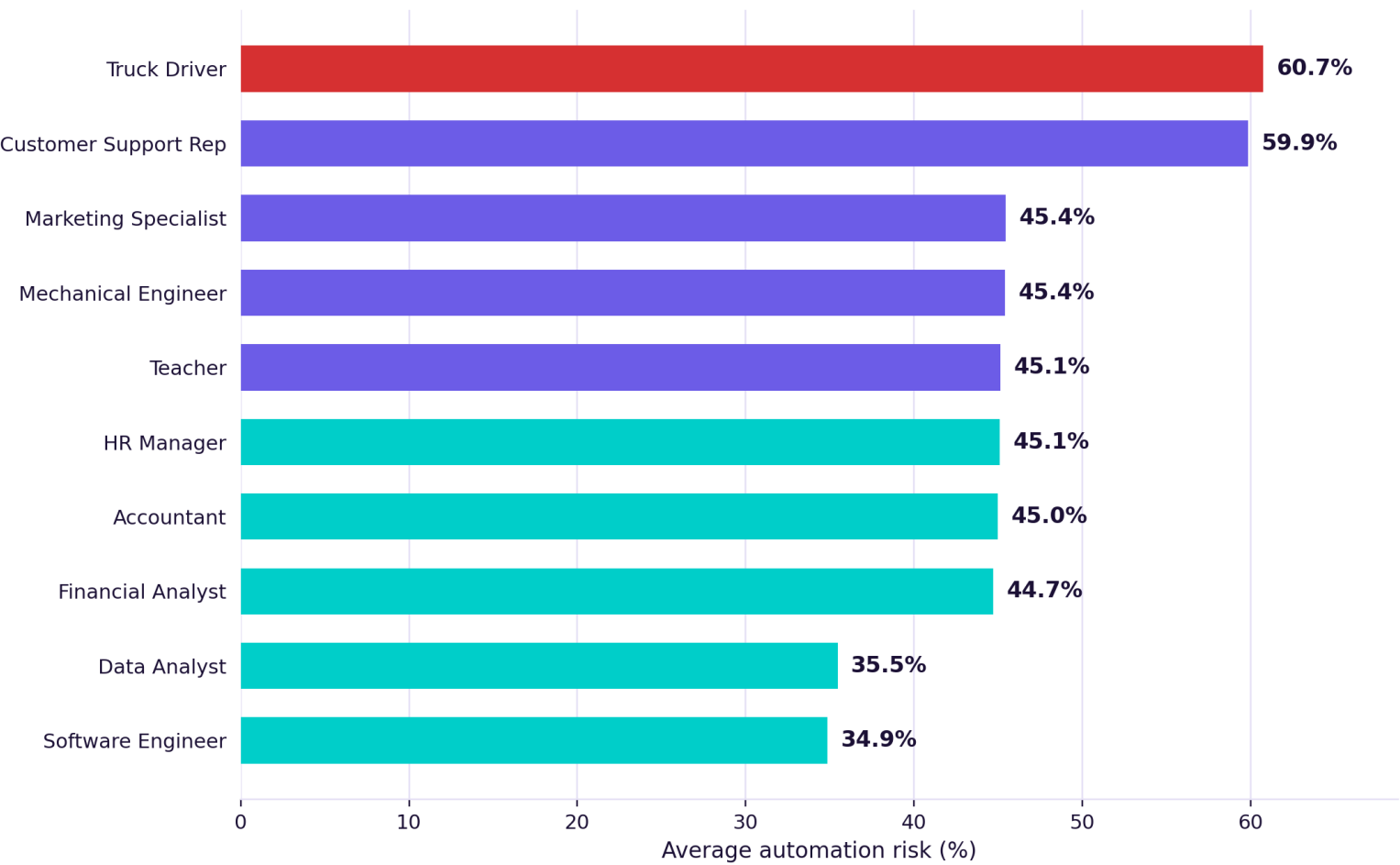
2020–2026

Period covered

Key variables: automation risk · AI adoption level · salary change · reskilling urgency · disruption intensity

QUESTION 1

Which occupations are most at risk of automation?



Higher risk

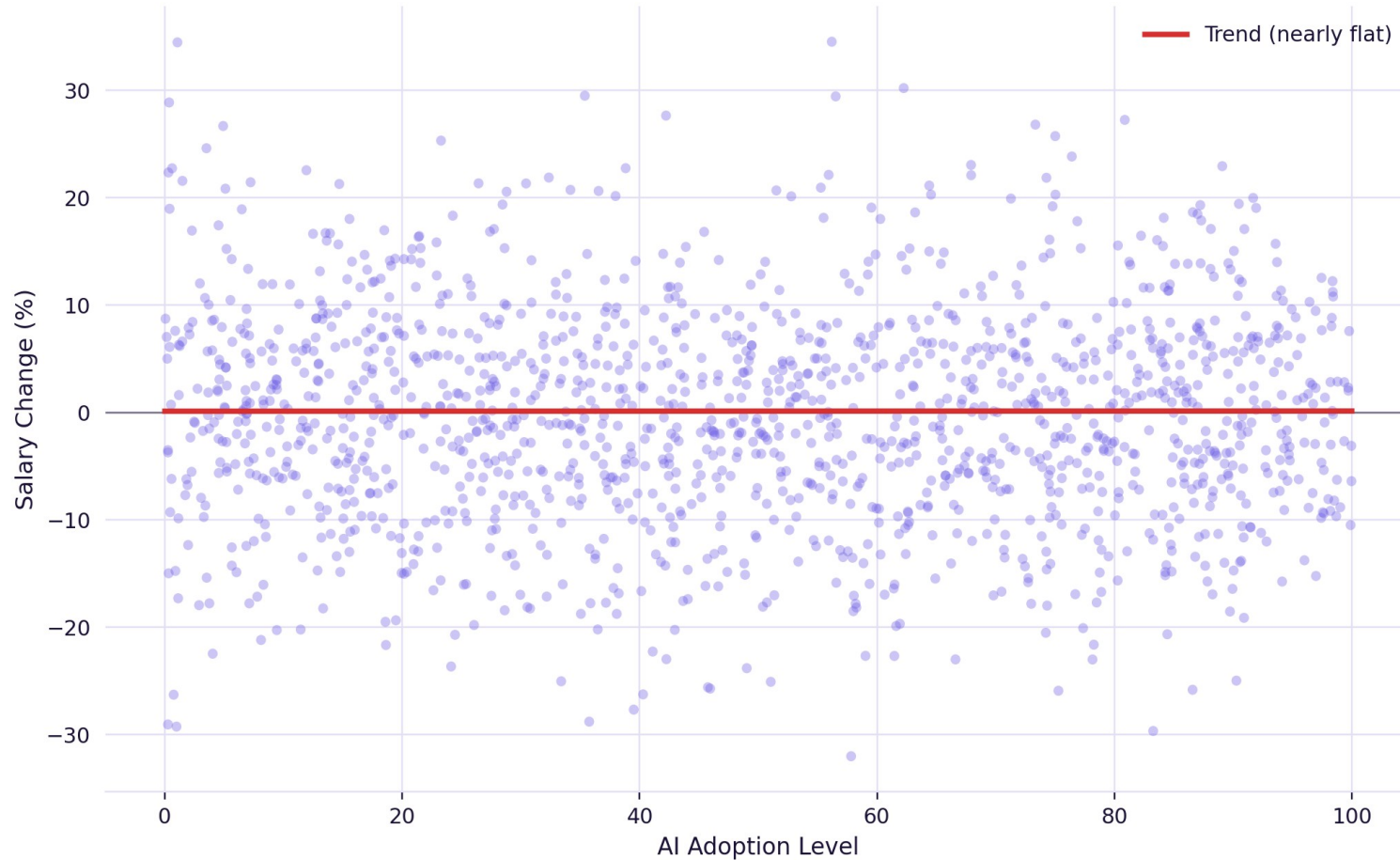
Truck Driver (60.7%) and Customer Support Rep (59.9%) concentrate the highest risks — repetitive, predictable tasks.

Lower risk

Software Engineer (34.9%) and Data Analyst (35.5%) are the safest — they require judgment and complex reasoning.

QUESTION 2

Is AI adoption linked to salary changes?



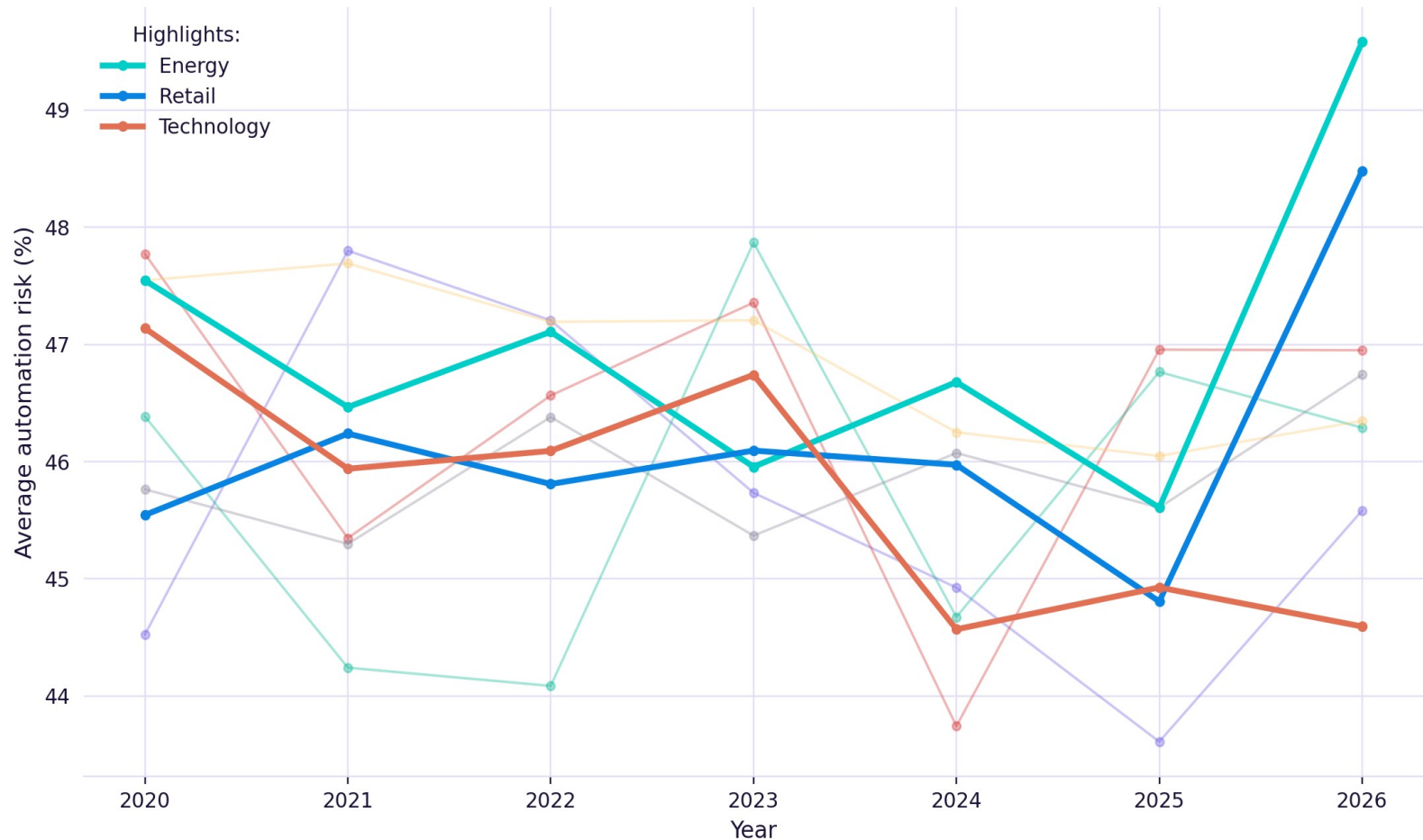
$$r \approx 0.000$$

PRACTICALLY NULL CORRELATION

In this dataset, adopting more AI is not associated with earning more or less. Salary seems to depend more on role and specialization than on AI exposure.

QUESTION 3

How has automation risk evolved (2020–2026)?



Overall: stable

Total change of only +0.36 p.p. over 6 years.

Technology: ↓ -2.5 p.p.

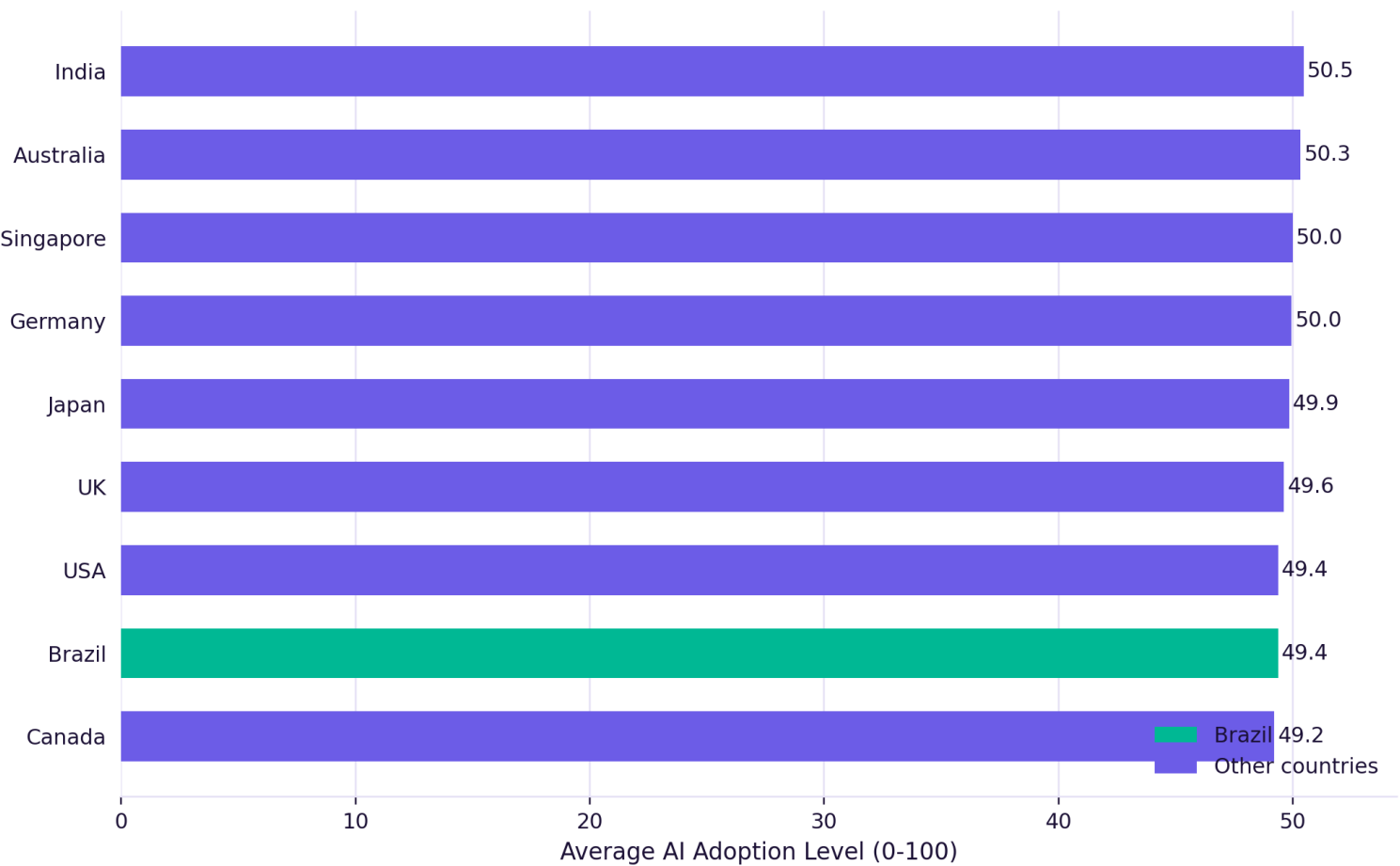
Professionals becoming AI 'operators'.

Retail: ↑ +3.0 p.p.

Automation advancing faster than adaptation.

QUESTION 4

How does Brazil compare to other countries?



BRAZIL

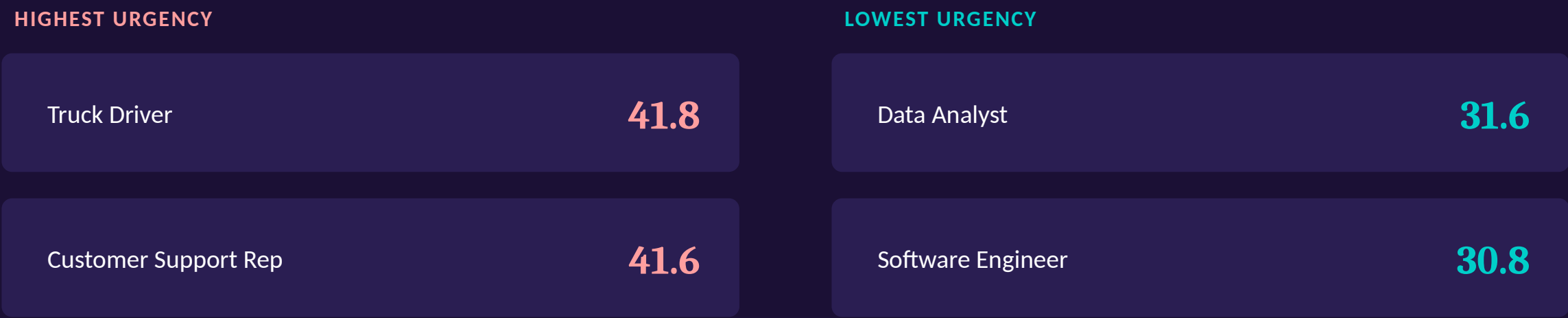
Brazil at the group average

- AI adoption: 49.4 (close to the overall average)
- Salary change: -0.06% (stable)
- Same level as USA (-0.05%) and UK (-0.11%)
- Germany (+0.58%) and India (+0.38%) lead in positive salary change

QUESTION 5

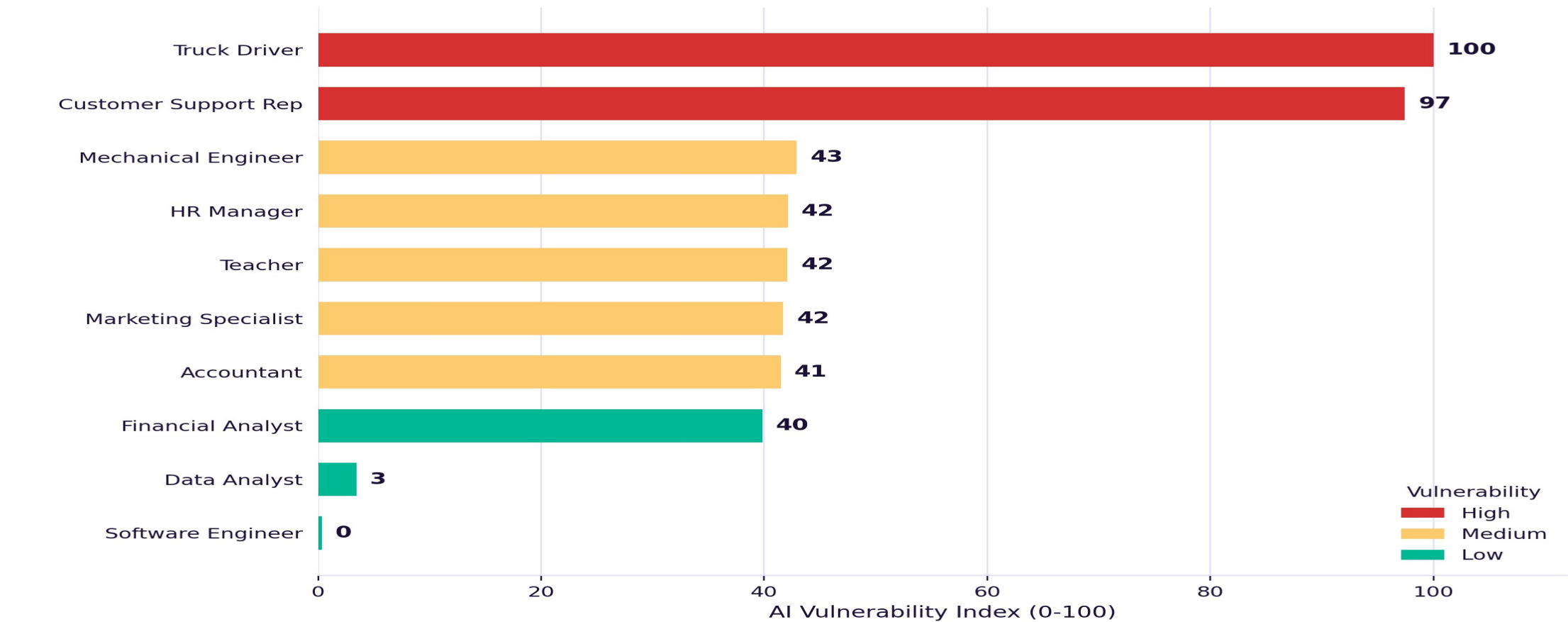
Who needs reskilling the most right now?

The reskilling urgency ranking closely follows the automation risk ranking — reinforcing the same pattern under a different metric.



AI Vulnerability Index (AIVI)

A custom metric, combining automation risk (40%), reskilling urgency (35%), and disruption intensity (25%) into a single index from 0 to 100.



What should you do with this information?

HIGH VULNERABILITY

Prioritize skills that complement AI: supervising automated systems, handling complex exceptions, transitioning to hybrid roles.

MEDIUM VULNERABILITY

Now is the time to become 'AI-fluent' in your own field — using AI tools as part of your workflow, not as a threat.

LOW VULNERABILITY

Keep evolving: technical fields change fast, and those who understand data and AI today tend to become increasingly relevant.

AI does not affect all occupations the same way.

The impact is much more tied to the type of task performed — repetitive vs. analytical/creative — than to sector or country. And, counterintuitively, higher AI adoption did not translate into greater salary gains or losses in this study.

Understanding this inequality through data — and translating it into an actionable index — is the first step toward better preparing for this transition.

FINAL REFLECTION

The question isn't whether AI will
change your job.

It's whether you've already
started preparing.

The data in this project shows one thing clearly: those who understand this transition — rather than fear it — come out ahead.

Thank you!